

EAST LONDON, South Africa

WMO: 688580

Lat: 33.0358S

Lon: 27.8161E

Elev: 381

StdP: 14.49

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 46.3	(c) 48.2	(d) 27.5	(e) 21.7	(f) 64.4	(g) 30.9	(h) 25.5	(i) 62.7	(j) 29.7	(k) 60.3	(l) 27.0	(m) 59.7	(n) 7.9	(o) 330	(p) 0.445

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 2	(b) 12.3	(c) 87.4	(d) 68.0	(e) 84.0	(f) 68.9	(g) 81.1	(h) 69.2	(i) 74.8	(j) 81.5	(k) 73.6	(l) 79.3	(m) 72.5	(n) 77.9	(o) 10.4	(p) 70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.0	124.3	78.2	71.7	118.8	76.8	70.4	113.6	75.7	38.6	81.9	37.4	79.6	36.4	77.8	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 27.0	(b) 23.8	(c) 21.3	(e) 41.1	(f) 96.2	(g) 2.3	(h) 4.5	(i) 39.5	(j) 99.5	(k) 38.1	(l) 102.1	(m) 36.8	(n) 104.7	(o) 35.1	(p) 108.0		
(a) 27.0	(b) 23.8	(c) 21.3	(e) 37.8	(f) 78.0	(g) 1.7	(h) 2.3	(i) 36.5	(j) 79.7	(k) 35.5	(l) 81.0	(m) 34.6	(n) 82.3	(o) 33.4	(p) 84.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	65.9	71.5	72.2	70.3	67.2	64.5	61.4	60.6	61.3	62.4	64.3	66.4	69.4
	DBStd	5.88	3.63	3.53	3.72	4.46	4.70	4.90	5.24	5.00	4.15	4.15	3.98	3.83
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	726	2	1	4	25	64	134	161	140	99	62	28	6
	CDD50	5814	665	623	630	516	450	341	329	349	371	445	492	603
	CDD65	1066	202	204	169	90	49	26	25	24	20	42	71	144
	CDH74	4351	749	831	618	348	280	196	211	207	103	146	197	465
	CDH80	882	115	133	109	81	85	54	62	78	24	31	36	74
Wind	WSAvg	10.5	10.8	10.5	10.0	9.4	9.3	9.8	9.8	10.8	11.1	11.7	11.6	11.1
Precipitation	PrecAvg	34.0	3.0	3.3	3.9	3.1	1.8	1.4	1.7	1.9	2.8	3.8	4.2	2.9
	PrecMax	53.1	9.1	9.0	23.6	12.8	6.3	7.7	8.7	8.8	10.2	12.4	15.2	9.6
	PrecMin	22.2	0.6	0.7	0.2	0.3	0.0	0.0	0.0	0.0	0.1	0.2	0.5	0.3
	PrecStd	7.6	1.9	2.0	3.6	2.5	1.7	1.8	2.2	1.7	2.1	2.8	3.1	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	87.9	88.0	87.9	88.3	88.2	86.0	86.4	89.5	83.8	84.0	84.0	86.4
		MCWB	74.2	75.3	72.9	69.4	64.2	58.8	59.4	60.5	63.6	66.0	70.0	72.1
	2%	DB	82.8	83.8	82.7	82.0	80.9	80.3	80.3	79.1	76.7	77.4	78.6	81.6
		MCWB	73.4	73.9	71.8	69.2	64.0	58.3	58.1	59.7	63.3	66.6	69.0	71.5
	5%	DB	80.4	80.9	79.8	77.7	75.8	74.1	73.8	73.5	73.0	73.8	75.5	78.8
		MCWB	71.9	72.5	70.5	67.7	62.0	57.1	57.0	59.1	62.4	65.1	67.2	70.4
	10%	DB	78.1	78.9	77.2	75.0	72.9	70.2	69.9	69.8	69.9	71.5	73.5	76.6
		MCWB	70.6	71.3	69.2	66.5	61.6	56.8	56.1	58.4	61.3	63.6	65.8	69.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.4	77.2	75.4	72.9	69.2	64.6	63.4	66.0	68.2	69.9	71.9	74.7
		MCDB	83.3	84.7	82.6	82.1	78.9	72.2	71.9	74.2	74.9	78.2	79.3	81.9
	2%	WB	74.4	75.2	73.3	70.8	66.8	62.1	61.5	63.8	65.7	67.9	70.1	72.8
		MCDB	80.4	81.9	79.7	78.3	74.6	69.7	69.8	70.5	71.4	74.4	75.8	78.3
	5%	WB	73.0	73.8	71.9	69.2	65.0	60.4	60.1	62.3	64.4	66.5	68.8	71.6
		MCDB	78.3	79.1	77.6	75.7	72.0	68.9	68.5	68.9	70.1	71.9	73.8	76.7
	10%	WB	71.9	72.5	70.5	67.6	63.3	59.0	58.7	60.9	62.9	65.1	67.4	70.2
		MCDB	76.8	77.4	75.8	73.0	70.2	67.9	67.1	67.2	68.3	70.0	72.0	75.0
Mean Daily Temperature Range	5% DB	MDBR	12.1	12.3	13.0	14.4	17.0	18.8	18.6	17.0	14.9	13.3	12.7	12.3
		MCDBR	15.6	15.4	17.0	19.2	23.3	24.7	26.2	24.8	20.6	17.8	16.2	16.0
		MCWBR	8.4	8.2	8.8	10.1	12.0	11.9	12.7	12.4	11.2	10.0	9.5	8.6
	5% WB	MCDBR	13.7	14.3	15.2	16.5	19.2	20.2	20.3	18.9	17.1	15.8	14.5	14.0
MCWBR		8.1	8.3	8.6	9.6	11.5	12.4	12.2	11.3	10.7	9.3	8.9	8.2	
Clear-Sky Solar Irradiance	taub	0.406	0.406	0.390	0.362	0.343	0.327	0.330	0.382	0.409	0.418	0.398	0.401	
	taud	2.424	2.434	2.457	2.503	2.525	2.536	2.521	2.373	2.296	2.314	2.409	2.429	
	Ebn at Noon	296	289	282	273	262	259	264	262	272	282	297	298	
	Edh at Noon	39	37	34	29	25	23	25	33	40	42	39	39	
All-Sky Solar Radiation	RadAvg	2020	1860	1571	1251	992	887	950	1170	1476	1709	1964	2063	
	RadStd	128	111	123	73	43	49	58	70	105	144	119	129	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page